Problem · Unit 5 · 10 points · 60 min · High difficulty / trap

P03 — Bead on a parabolic wire spinning about its axis: Coriolis side force, Jacobi integral and the critical rate

*A smooth rigid wire lies in a vertical plane and has the shape of the parabola $y = x^2/(2p)$ ($p > 0$), with its vertex at O and its axis along the upward vertical. The wire is driven by a motor so that it spins about that vertical axis with **constant** angular rate $\Omega > 0$. A bead P of mass m is threaded on the wire.*

Let $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ be inertial ($\mathbf{j}_0$ vertical upward, gravity $-g\,\mathbf{j}_0$) and $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ be attached to the wire, with $\mathbf{j}_1 = \mathbf{j}_0$, the wire in the plane x_1y_1 and $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_0$. At $t = 0$ both frames coincide, so $\mathbf{i}_1$ has turned the angle $\theta = \Omega t$ about $\mathbf{j}_0$. The position of P on the wire is given by its abscissa x along $\mathbf{i}_1$ (positive or negative).

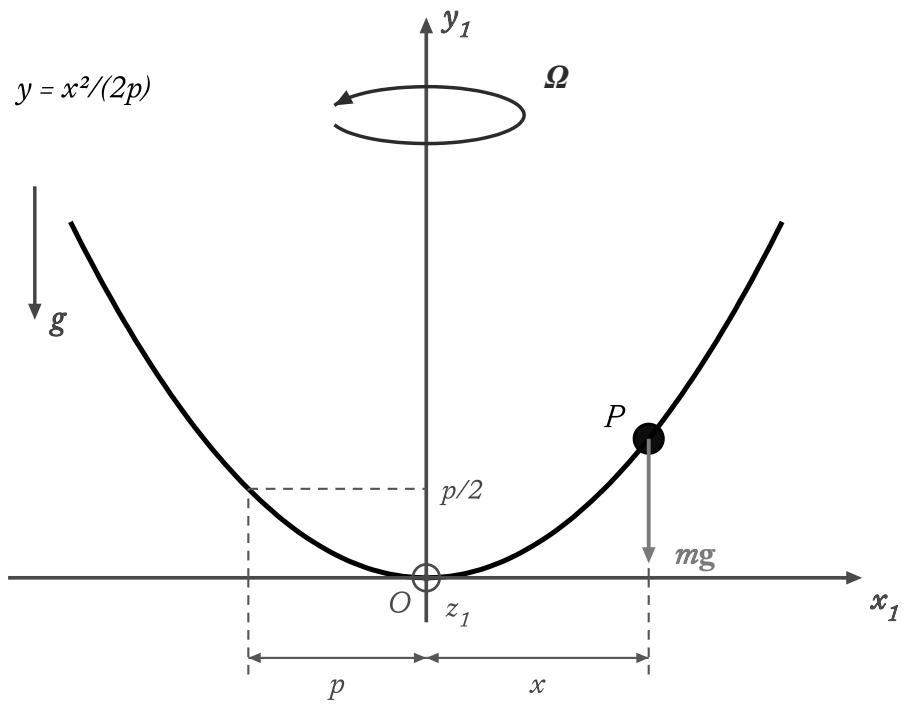


Figure 1. Wire $y = x^2/(2p)$ in the plane x_1y_1 of S_1, which turns with $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$ about the vertical axis; $\mathbf{k}_1$ is normal to the plane of the wire, gravity $-g\,\mathbf{j}$.

Setup

- Data: m, p, g, Ω. In part (g): $\Omega^2 = 2g/p$ and the bead is released **at rest relative to the wire** at $x_0 = p/2$.
- Hypotheses: point particle, no friction, bead threaded on the wire (bilateral constraint), S_0 inertial.
- Write $q = \sqrt{1 + x^2/p^2}$.

Part (a) · 1 point

Degrees of freedom and frames

Count the degrees of freedom and classify the constraints in S_0 and in S_1. Write the change-of-basis matrix $[{}_0R_1]$, check it, and give $\boldsymbol{\omega}_{10}$ and $\boldsymbol{\alpha}_{10}$.

Step 1. Count

In S_1 the wire is fixed: $z_1 = 0$ and $y_1 - x_1^2/(2p) = 0$ are two holonomic equations without time, so $3 - 2 = 1$ degree of freedom, coordinate x (Notes §2.7). Seen from S_0 the same curve moves, so the constraints contain t through $\theta = \Omega t$: rheonomic. This is why we work in S_1.

Step 2. Change of basis

The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$ (Notes §1.5): $\mathbf{i}_1 = \cos\theta\,\mathbf{i}_0 - \sin\theta\,\mathbf{k}_0$, $\mathbf{j}_1 = \mathbf{j}_0$, $\mathbf{k}_1 = \sin\theta\,\mathbf{i}_0 + \cos\theta\,\mathbf{k}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

Step 3. Angular velocity

$\dot{\mathbf{i}}_1 = \dot\theta(-\sin\theta\,\mathbf{i}_0 - \cos\theta\,\mathbf{k}_0) = -\Omega\,\mathbf{k}_1$ and $\boldsymbol{\omega} \times \mathbf{i}_1 = \Omega\,\mathbf{j}_1 \times \mathbf{i}_1 = -\Omega\,\mathbf{k}_1$ (Poisson, Notes §1.7), so $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{j}_0 = \Omega\,\mathbf{j}_0$, and $\boldsymbol{\alpha}_{10} = \dot{\boldsymbol{\omega}}_{10} = \mathbf{0}$ because Ω is constant.

KEY RESULTS

- 1 degree of freedom, coordinate x. Two holonomic constraints, scleronomic in S_1 ($z_1 = 0, y_1 = x_1^2/(2p)$), rheonomic in S_0.
- $[{}_0R_1] = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$, $\det = 1$, $RR^T = I$; $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_0$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$.

Part (b) · 1.5 points

Relative and absolute kinematics

Write $\mathbf{r}$, $\mathbf{v}_{rel}$, $\mathbf{a}_{rel}$ in $\mathcal{B}_1$, and the absolute acceleration of P with the Coriolis theorem.

Step 1. Relative motion

$\mathbf{r} = x\,\mathbf{i}_1 + \dfrac{x^2}{2p}\mathbf{j}_1$. Differentiating in S_1 (where $\mathbf{i}_1, \mathbf{j}_1$ are constant), with the chain rule $\dfrac{d}{dt}\dfrac{x^2}{2p} = \dfrac{x\dot x}{p}$ and the product rule $\dfrac{d}{dt}\dfrac{x\dot x}{p} = \dfrac{\dot x^2 + x\ddot x}{p}$:

$$\mathbf{v}_{rel} = \dot{x}\,\mathbf{i}_1 + \frac{x\dot{x}}{p}\mathbf{j}_1, \qquad \mathbf{a}_{rel} = \ddot{x}\,\mathbf{i}_1 + \frac{x\ddot{x} + \dot{x}^2}{p}\mathbf{j}_1$$

Step 2. Transport and Coriolis terms

O is fixed and $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so (Notes §3.2) $\mathbf{a} = \mathbf{a}_{rel} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + 2\boldsymbol{\omega} \times \mathbf{v}_{rel}$. With $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1$ and $\mathbf{j}_1 \times \mathbf{j}_1 = \mathbf{0}$:

$$\boldsymbol{\omega} \times \mathbf{r} = -\Omega x\,\mathbf{k}_1, \qquad \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = \Omega\,\mathbf{j}_1 \times (-\Omega x\,\mathbf{k}_1) = -\Omega^2 x\,\mathbf{i}_1, \qquad 2\boldsymbol{\omega} \times \mathbf{v}_{rel} = -2\Omega\dot{x}\,\mathbf{k}_1$$

$$\mathbf{a} = \big(\ddot{x} - \Omega^2 x\big)\mathbf{i}_1 + \frac{x\ddot{x} + \dot{x}^2}{p}\mathbf{j}_1 - 2\Omega\dot{x}\,\mathbf{k}_1$$

KEY RESULTS

- $\mathbf{v}_{rel} = \dot{x}\Big(\mathbf{i}_1 + \dfrac{x}{p}\mathbf{j}_1\Big)$, $\mathbf{a}_{rel} = \ddot{x}\Big(\mathbf{i}_1 + \dfrac{x}{p}\mathbf{j}_1\Big) + \dfrac{\dot{x}^2}{p}\mathbf{j}_1$.
- $\mathbf{a}_{tr} = -\Omega^2 x\,\mathbf{i}_1$, $\mathbf{a}_{cor} = -2\Omega\dot{x}\,\mathbf{k}_1$, $\mathbf{a} = \mathbf{a}_{rel} + \mathbf{a}_{tr} + \mathbf{a}_{cor}$.

Part (c) · 1.5 points

Forces and inertia forces

Write the applied forces, the reaction of the wire and the inertia forces of S_1 in $\mathcal{B}_1$. Which of them do work in S_1?

Step 1. Applied force and reaction

$\mathbf{W} = -mg\,\mathbf{j}_1$. The tangent of the wire is along $\boldsymbol{\tau} = \mathbf{i}_1 + \dfrac{x}{p}\mathbf{j}_1$; the normal plane is spanned by the in-plane normal $\mathbf{n} = \dfrac{1}{q}\Big(-\dfrac{x}{p}\mathbf{i}_1 + \mathbf{j}_1\Big)$ and by $\mathbf{k}_1$. A smooth wire gives (Notes §6.5) $\mathbf{N} = N_n\,\mathbf{n} + N_z\,\mathbf{k}_1$, two unknowns.

Step 2. Inertia forces

In S_1 Newton's law reads $m\mathbf{a}_{rel} = \mathbf{W} + \mathbf{N} + \mathbf{F}_{cf} + \mathbf{F}_{cor}$ (Notes §4.9), with

$$\mathbf{F}_{cf} = -m\,\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = m\Omega^2 x\,\mathbf{i}_1, \qquad \mathbf{F}_{cor} = -2m\,\boldsymbol{\omega} \times \mathbf{v}_{rel} = 2m\Omega\dot{x}\,\mathbf{k}_1$$

There is no Euler term ($\boldsymbol{\alpha} = \mathbf{0}$) and no translational term (O fixed).

Step 3. Work in S_1

$\mathbf{N} \cdot \mathbf{v}_{rel} = 0$ and $\mathbf{F}_{cor} \cdot \mathbf{v}_{rel} = 0$ (both orthogonal to the tangent). $\mathbf{W}$ has potential $mgy = \dfrac{mgx^2}{2p}$ and $\mathbf{F}_{cf}$ has potential $V_{cf} = -\tfrac{1}{2}m\Omega^2x^2$ (check: $-\dfrac{dV_{cf}}{dx} = m\Omega^2 x$).

KEY RESULTS

- $\mathbf{W} = -mg\,\mathbf{j}_1$; $\mathbf{N} = N_n\mathbf{n} + N_z\mathbf{k}_1$ with $\mathbf{n} = \dfrac{-(x/p)\mathbf{i}_1 + \mathbf{j}_1}{q}$.
- $\mathbf{F}_{cf} = m\Omega^2 x\,\mathbf{i}_1$ (potential $-\tfrac{1}{2}m\Omega^2x^2$), $\mathbf{F}_{cor} = 2m\Omega\dot{x}\,\mathbf{k}_1$ (no work in S_1).

Part (d) · 2 points

Equation of motion and reactions

Obtain the equation of motion free of reactions and the two components of the reaction.

Step 1. Tangential projection

Dot $m\mathbf{a}_{rel} = \mathbf{W} + \mathbf{N} + \mathbf{F}_{cf} + \mathbf{F}_{cor}$ with $\boldsymbol{\tau} = \mathbf{i}_1 + \dfrac{x}{p}\mathbf{j}_1$. Left: $m\left[\ddot{x} + \dfrac{x}{p}\cdot\dfrac{x\ddot{x} + \dot{x}^2}{p}\right]$. Right: $\mathbf{W}\cdot\boldsymbol{\tau} = -\dfrac{mgx}{p}$, $\mathbf{F}_{cf}\cdot\boldsymbol{\tau} = m\Omega^2 x$, and $\mathbf{N}\cdot\boldsymbol{\tau} = \mathbf{F}_{cor}\cdot\boldsymbol{\tau} = 0$:

$$\left(1 + \frac{x^2}{p^2}\right)\ddot{x} + \frac{x}{p^2}\dot{x}^2 = x\left(\Omega^2 - \frac{g}{p}\right)$$

Step 2. Out-of-plane projection

$\mathbf{a}_{rel}\cdot\mathbf{k}_1 = 0$, so $0 = N_z + 2m\Omega\dot{x}$:

$$N_z = -2m\Omega\dot{x}$$

The wire must cancel the Coriolis force. Seen from S_0: a bead moving outwards must gain the azimuthal speed Ωx of the wire; the drag direction of the rotation at P is $\boldsymbol{\omega} \times \mathbf{r} = -\Omega x\, \mathbf{k}_1$, and for $x > 0, \dot{x} > 0$ the wire indeed pushes along $-\mathbf{k}_1$.

Step 3. In-plane normal projection

Dot with $q\,\mathbf{n} = -\dfrac{x}{p}\mathbf{i}_1 + \mathbf{j}_1$. Left: $m\Big[-\dfrac{x}{p}\ddot{x} + \dfrac{x\ddot{x} + \dot{x}^2}{p}\Big] = \dfrac{m\dot{x}^2}{p}$. Right: $-mg$ (weight) $-\dfrac{m\Omega^2 x^2}{p}$ (centrifugal) $+qN_n$:

$$N_n = \frac{m}{q}\Big(\frac{\dot{x}^2}{p} + \frac{\Omega^2 x^2}{p} + g\Big)$$

a sum of positive terms: the wire always pushes the bead towards the concave side.

Step 4. Cross-check in S_0

As an extension (Lagrange equations are not in the course notes), the Lagrange equation of $L = \frac{1}{2}m|\mathbf{v}|^2 - mgy$ with the absolute velocity $\mathbf{v} = \mathbf{v}_{rel} - \Omega x\, \mathbf{k}_1$, $|\mathbf{v}|^2 = q^2\dot{x}^2 + \Omega^2 x^2$, gives the same equation (checked with SymPy).

KEY RESULTS

- $\Big(1 + \dfrac{x^2}{p^2}\Big)\ddot{x} + \dfrac{x}{p^2}\dot{x}^2 = x\Big(\Omega^2 - \dfrac{g}{p}\Big)$.
- $N_z = -2m\Omega\dot{x};\ N_n = \dfrac{m}{q}\Big(\dfrac{\dot{x}^2}{p} + \dfrac{\Omega^2 x^2}{p} + g\Big) > 0.$

Part (e) · 1 point

Jacobi first integral

Obtain a first integral of the motion. Is it the mechanical energy of the bead in S_0?

Step 1. Exact derivatives

By the product and chain rules, $\dfrac{d}{dt}\Big[\frac{1}{2}\Big(1 + \dfrac{x^2}{p^2}\Big)\dot{x}^2\Big] = \Big(1 + \dfrac{x^2}{p^2}\Big)\dot{x}\ddot{x} + \dfrac{x\dot{x}}{p^2}\dot{x}^2 = \dot{x}\Big[\Big(1 + \dfrac{x^2}{p^2}\Big)\ddot{x} + \dfrac{x}{p^2}\dot{x}^2\Big]$, and $\dot{x}\cdot x\Big(\Omega^2 - \dfrac{g}{p}\Big) = \dfrac{d}{dt}\Big[\frac{1}{2}\Big(\Omega^2 - \dfrac{g}{p}\Big)x^2\Big]$. Integrating from 0 to t (Barrow's rule):

$$J = \frac{1}{2}\Big(1 + \frac{x^2}{p^2}\Big)\dot{x}^2 + \frac{1}{2}\Big(\frac{g}{p} - \Omega^2\Big)x^2 = J(0)$$

Step 2. Meaning

$\frac{1}{2}|\mathbf{v}_{rel}|^2 = \frac{1}{2}q^2\dot{x}^2$ and the potentials of (c): $J = \frac{1}{2}|\mathbf{v}_{rel}|^2 + gy - \frac{1}{2}\Omega^2x^2$. In S_1 only $\mathbf{W}$ and $\mathbf{F}_{cf}$ work, so this "relative energy" is conserved. In S_0, $\mathbf{N}\cdot\mathbf{v} = N_z\mathbf{k}_1\cdot(-\Omega x\,\mathbf{k}_1) = 2m\Omega^2x\dot{x} \neq 0$: the motor feeds energy to the bead.

KEY RESULTS

- $J = \frac{1}{2}\left(1+\frac{x^2}{p^2}\right)\dot{x}^2 + \frac{1}{2}\left(\frac{g}{p}-\Omega^2\right)x^2 = \text{const}$ (per unit mass).
- It is the energy **in** S_1 including the centrifugal potential; the energy in S_0 is not conserved (the motor works through N_z).

Part (f) · 1.5 points

Relative equilibrium and the critical rate

Find the positions of relative equilibrium and their stability for $\Omega^2 < g/p$, $\Omega^2 = g/p$ and $\Omega^2 > g/p$. Give the frequency of small oscillations when it exists. What happens at the critical rate?

Step 1. Equilibria

$\dot{x} = \ddot{x} = 0$ in (d) gives $x(\Omega^2 - g/p) = 0$. If $\Omega^2 \neq g/p$ only $x = 0$ (the vertex). $V''_{eff}(0) = m(g/p - \Omega^2)$ (Notes §4.6): a minimum (stable) for $\Omega^2 < g/p$, a maximum (unstable) for $\Omega^2 > g/p$.

Step 2. Small oscillations

Linearising (d) about $x = 0$ ($q \to 1$, the $x\dot{x}^2$ term is of third order): $\ddot{x} + \left(\frac{g}{p} - \Omega^2\right)x = 0$, so $\omega = \sqrt{g/p - \Omega^2}$ for $\Omega^2 < g/p$.

Step 3. The critical rate (the trap)

For $\Omega^2 = g/p$ the linearised equation is $\ddot{x} = 0$, which does not decide anything by itself. The full equation becomes $q^2\ddot{x} + \frac{x}{p^2}\dot{x}^2 = 0$, so $\ddot{x} = 0$ whenever $\dot{x} = 0$, **at any** x: every point is a relative equilibrium. J reduces to $\frac{1}{2}q^2\dot{x}^2 = |\mathbf{v}_{rel}|^2/2 = \text{const}$: the bead slides with constant relative speed and never comes back (neutral, not stable). Geometric reason: at relative rest the resultant of gravity and centrifugal force must be normal to the wire, i.e. $dy/dx = \Omega^2x/g$; the parabola has $dy/dx = x/p$, which matches at every x exactly when $\Omega^2 = g/p$ (it is the shape of the free surface of a liquid spinning at that rate).

KEY RESULTS

- $x = 0$ is an equilibrium for every Ω: stable for $\Omega^2 < g/p$ ($\omega^2 = g/p - \Omega^2$), unstable for $\Omega^2 > g/p$.
- $\Omega^2 = g/p$: **every** point of the wire is an equilibrium (neutral), and a bead with relative speed slides at constant speed along the wire: $q^2\dot{x}^2 = \text{const}$.

Part (g) · 1.5 points

Release at $\Omega^2 = 2g/p$: escape and side force

For $\Omega^2 = 2g/p$ and release from relative rest at $x_0 = p/2$: show that the bead moves outwards, give $\dot{x}^2(x)$ and $N_z(x)$, their values at $x = p$ and their limits for $x \to \infty$.

Step 1. Direction

With $\dot{x} = 0$ in (d): $q_0^2 \ddot{x} = x_0\left(\frac{2g}{p} - \frac{g}{p}\right) = \frac{g x_0}{p} > 0$. The bead starts outwards (the vertex is unstable, part (f)).

Step 2. Speed

$J(x, \dot{x}) = J(x_0, 0)$ with $g/p - \Omega^2 = -g/p$:

$$\frac{1}{2}\left(1 + \frac{x^2}{p^2}\right)\dot{x}^2 - \frac{g}{2p}x^2 = -\frac{g}{2p}x_0^2 \implies \dot{x}^2 = \frac{g}{p}\,\frac{x^2 - x_0^2}{1 + x^2/p^2}$$

$\dot{x}^2 \geq 0$ only for $|x| \geq x_0$: the bead never returns towards the vertex, and x grows monotonically.
Consistency: $\frac{d(\dot{x}^2)}{dx} = 2\ddot{x}$ reproduces (d) (checked with SymPy).

Step 3. Side force

$N_z = -2m\Omega\dot{x}$ with $\Omega = \sqrt{2g/p}$ and $\dot{x} > 0$:

$$N_z = -2m\sqrt{\frac{2g}{p}}\sqrt{\frac{g}{p}\,\frac{x^2 - x_0^2}{1 + x^2/p^2}} = -\frac{2\sqrt{2}\,mg}{p}\sqrt{\frac{x^2 - x_0^2}{1 + x^2/p^2}}$$

Step 4. Values

At $x = p$ ($x_0 = p/2$): $\dot{x}^2 = \frac{g}{p}\cdot\frac{p^2 - p^2/4}{2} = \frac{3}{8}gp$, and $N_z = -2m\sqrt{\frac{2g}{p}}\sqrt{\frac{3gp}{8}} = -2mg\sqrt{\frac{3}{4}} = -\sqrt{3}\,mg$.
The in-plane reaction there: $\frac{\dot{x}^2}{p} + \frac{\Omega^2 x^2}{p} + g = \frac{3}{8}g + 2g + g = \frac{27}{8}g$ and $q = \sqrt{2}$, so $N_n = \frac{27}{8\sqrt{2}}mg \approx 2.386\,mg$. The relative speed is then $|\mathbf{v}_{rel}|^2 = q^2\dot{x}^2 = \frac{3}{4}gp$.

Step 5. Limits

For $x \to \infty$, $\frac{x^2 - x_0^2}{1 + x^2/p^2} \to p^2$, so $\dot{x}^2 \to gp$ and $|N_z| \to 2\sqrt{2}\,mg$. The abscissa grows at a bounded rate, while the speed along the wire $q|\dot{x}|$ grows without bound because the wire becomes steep.

KEY RESULTS

- $\ddot{x}(x_0) = \dfrac{(g/p)\,x_0}{1 + x_0^2/p^2} > 0;\ \dot{x}^2 = \dfrac{g}{p}\,\dfrac{x^2 - x_0^2}{1 + x^2/p^2}.$
- $N_z = -\dfrac{2\sqrt{2}\,mg}{p}\sqrt{\dfrac{x^2 - x_0^2}{1 + x^2/p^2}}$; at $x = p$: $\dot{x}^2 = \tfrac{3}{8}gp$, $N_z = -\sqrt{3}\,mg \approx -1.732\,mg$, $N_n = \dfrac{27}{8\sqrt{2}}mg \approx 2.386\,mg$.
- $x \to \infty$: $\dot{x}^2 \to gp$ and $|N_z| \to 2\sqrt{2}\,mg \approx 2.83\,mg$.

Interpretation and checks

- **Dimensional check:** g/p and Ω^2 are s^{-2}; in (d) every term is $\mathrm{m\ s^{-2}}$ ($x\dot{x}^2/p^2$ is $\mathrm{m\ (m/s)^2/m^2}$); $N_z = 2m\Omega\dot{x}$ is $\mathrm{kg\ s^{-1}\ m\ s^{-1}}$ = N; $\dot{x}^2 \to gp$ is $\mathrm{m^2\ s^{-2}}$.
- **Limits:** $\Omega = 0$: a bead in a fixed parabolic bowl, $\omega = \sqrt{g/p}$ about the vertex and $N_z = 0$; $p \to \infty$ (straight horizontal wire): always $\Omega^2 > g/p$, and $x(t)$ grows as $\cosh \Omega t$, the classic rotating rod.
- **Common mistakes in the exam:** (1) forgetting N_z (the reaction of a wire has two components; the Coriolis force needs one of them); (2) writing the Jacobi integral as the absolute energy (the motor does work in S_0); (3) at $\Omega^2 = g/p$, concluding "stable" or "unstable" from $\ddot{x} = 0$: every point is an equilibrium and a moving bead keeps a constant relative speed; (4) dropping the $(x/p^2)\dot{x}^2$ term, which comes from differentiating the constraint $y = x^2/(2p)$ twice.